



INDIAN INSTITUTE OF TECHNOLOGY KANPUR

Project Report

Self-Optimal Clustering using Threshold Function

Course: **EE656 AI, ML, DL and It's Applications**
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Abstract

Self-optimal clustering (SOC) serves as an enhanced variation of the Improved Mountain Clustering (IMC) methodology, which was initially formulated by **Dr. Nishchal Kumar Verma** and **Abhishek Roy**. The effectiveness of SOC is measured by observing the internal compactness of the clusters alongside their separation from one another. By utilizing interpolation, the optimizing factor within the threshold function is calculated. This mathematical adjustment proves highly effective in generating superior quality clusters, a fact confirmed through visual analysis and standard metrics like the global silhouette index, partition index, and separation index. Consequently, SOC stands as an efficient and dependable technique for fields such as machine learning, data mining, and pattern recognition.

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1 Introduction to Clustering

(1.1) What is Clustering?

Clustering can be compared to organizing a large assortment of candies into separate bowls based on their type, such as chocolates in one bowl and sour candies in another. In data terms, clustering is the act of grouping similar items together.

The primary objectives of forming a cluster are:

- To ensure that elements within the same group share a high degree of similarity.
- To guarantee that elements across different groups remain distinctly different.

Two major types of Clustering:

- **Hard Clustering:** Every individual item is assigned to strictly one group. For example, assigning a student to a single specific classroom.
- **Fuzzy Clustering:** Items have the flexibility to belong to multiple groups simultaneously, with varying degrees of association. For instance, a person might belong to a sports group at 80%, a music group at 60%, and a reading group at 40%. Fuzzy clustering can also be translated into hard clustering by simply assigning the item to the group where it holds the highest score.

(1.2) A Few Popular Methods

1. **Fuzzy C-Means (FCM):** Allows data points to have membership in multiple clusters. While it yields reliable results, it can be computationally time-consuming.
2. **Probabilistic Clustering:** Assigns items to a single cluster based on calculated probabilities. It offers precision but requires significant processing time.
3. **Mountain Clustering:** Maps data onto a conceptual mountain topology where the highest peaks represent potential cluster centers. However, earlier iterations occasionally overlooked optimal clusters by flattening nearby data points too rapidly.
4. **Improved Mountain Clustering (IMC):** An upgraded iteration that continuously searches for valid clusters. It operates faster and delivers more precise groupings.

(1.3) What this Paper Proposes: SOC (Self-Optimal Clustering)

The authors introduce Self-Optimal Clustering (SOC), an innovative methodology with several advantages:

- It autonomously identifies the optimal number of groups.
- It integrates mathematical interpolation to boost precision.
- It surpasses legacy methods in both speed and clustering quality.
- It demonstrates high adaptability across diverse datasets.

2 Overview of Clustering Techniques

(2.1) Overview

This segment provides a brief overview of the diverse clustering techniques researchers have engineered over the years. Similar to organizing a closet by size, color, or type, data can be grouped in numerous ways based on the desired outcome.

(2.2) Different Approaches to Clustering

Engineers and scientists have explored multiple intelligent strategies for data grouping:

- **Local variations:** Grouping items based on localized patterns, akin to grouping neighborhoods by the characteristics of nearby houses.
- **Normalized cuts:** Dividing data based on similarities, similar to slicing a cake into balanced portions.
- **Feature analysis:** Concentrating on key traits, such as organizing students by their preferred subjects.
- **Saddle point detection:** Identifying peaks or valleys in data to demarcate where one segment ends and another begins.
- **Multiresolution:** Analyzing data by zooming in for granular details and zooming out for broader perspectives.
- **Color-texture regions:** Primarily utilized in image processing, this involves categorizing items by visual feel and color.
- **Stochastic clustering:** Utilizing randomness to evaluate various grouping configurations to find the most effective one.

(2.3) How Is SOC Better?

The SOC method relies on a mathematical tool known as Lagrange Interpolation to pinpoint the optimal thresholds for clustering.

- SOC acts as an advanced upgrade to IMC.
- It determines the ideal data groupings through mathematical threshold adjustments rather than estimation.
- By leveraging Lagrange interpolation, it utilizes known data points to make accurate predictions, significantly enhancing clustering precision.

3 Self-Optimal Clustering: SOC

(3.1) Algorithm

Step 1: Normalization

Normalize the data across every dimension in hyperspace to bound the data points within a unit hypercube. Let the j -th instance of data in x hyperspace be defined as:

$$x_j = \begin{pmatrix} x_j^1 \\ x_j^2 \\ \vdots \\ x_j^D \end{pmatrix}$$

where D represents the total number of dimensions. The normalized instance $\bar{x}_j$ is defined as

$$\bar{x}_j = \frac{x_j - x_{min}}{x_{max} - x_{min}}, \forall j = 1, 2, \dots, n$$

where n is the total number of data points.

Step 2: Threshold Computation

Calculate the threshold value δ_m , a positive metric that establishes the neighborhood of data points for the m -th cluster. This value δ_m is derived from a heuristic expression multiplied by an optimizing factor β_m . Initially, β_m is set to 1 when evaluating the m -th cluster for the very first time. The threshold function is formulated as

$$\delta_m = \left(\frac{1}{2n} \sum_{j=1}^n \min \left(\sum_{i=1}^D x_j^i \right) \right) \cdot \beta_m$$

Step 3: Potential Function Computation

Determine the potential value P_r^m for each data point $\bar{x}_r$ by utilizing the mountain function, which relies on the squared Euclidean distance between x_r and all other normalized points $\bar{x}_j$.

$$d^2(\bar{x}_r, \bar{x}_j) = (\bar{x}_r - \bar{x}_j)^T Q (\bar{x}_r - \bar{x}_j)$$

where Q is the identity matrix ($Q = I$). The potential function is then calculated as:

$$P_r^m = \sum_{j=1}^n \exp \left(-\frac{d^2(\bar{x}_r, \bar{x}_j)}{\delta_m^2} \right)$$

Step 4: Selection of cluster centers

Identify the data point that corresponds to the highest potential value among $P_1^m, P_2^m, \dots, P_n^m$ to serve as the m -th cluster center $\bar{c}_m$.

$$\bar{c}_m = \bar{x}^* \Leftrightarrow P_m^* = \max_{r=1}^n (P_r^m)$$

Step 5: Cluster Assignment

Allocate data points to the m -th cluster if their Euclidean distance to the center $\bar{c}_m$ is less than or equal to the threshold δ_m .

$$d^2(\bar{x}_r, \bar{c}_m) \leq \delta_m, \forall r = 1, 2, \dots, n$$

Step 6: Eliminate all data points assigned to the current m -th cluster from the dataset.

Step 7: Iterate through Steps 2 to 6 on the reduced dataset to form subsequent clusters, up to the optimal number of clusters M .

Step 8: Distribution of Remaining Data Points Assign the remaining data points to the already formed clusters based on their Euclidean distances from the corresponding cluster centers. Each point is assigned to the nearest cluster center.

Step 9: Calculate the Global Silhouette Index (GSI) Evaluate the quality of the obtained clusters using the Global Silhouette Index (GSI). A GSI value close to unity indicates better cluster formation. This is achieved when the silhouette values S_m for all clusters tend towards unity.

Let the threshold value and silhouette value corresponding to the t^{th} cluster be denoted by δ_t and S_t , respectively.

Step 10: Interpolation of Threshold vs. Silhouette Value

Obtain the relationship between the threshold values and the corresponding silhouette values using Lagrange interpolation.

For a total of M clusters, the known pairs are

$$(\delta_1, S_1), (\delta_2, S_2), \dots, (\delta_M, S_M).$$

The interpolation polynomial is

$$S_t = \sum_{m=1}^M S_m \ell_m(\delta_t), \quad (1)$$

where the Lagrange basis polynomial is

$$\ell_m(\delta_t) = \prod_{\substack{k=1 \\ k \neq m}}^M \frac{\delta_t - \delta_k}{\delta_m - \delta_k}. \quad (2)$$

Substituting the basis polynomial gives

$$S_t = \sum_{m=1}^M S_m \prod_{\substack{k=1 \\ k \neq m}}^M \frac{\delta_t - \delta_k}{\delta_m - \delta_k}. \quad (3)$$

Expanding the expression,

$$\begin{aligned} S_t &= S_1 \frac{(\delta_t - \delta_2)(\delta_t - \delta_3) \cdots (\delta_t - \delta_M)}{(\delta_1 - \delta_2)(\delta_1 - \delta_3) \cdots (\delta_1 - \delta_M)} \\ &\quad + S_2 \frac{(\delta_t - \delta_1)(\delta_t - \delta_3) \cdots (\delta_t - \delta_M)}{(\delta_2 - \delta_1)(\delta_2 - \delta_3) \cdots (\delta_2 - \delta_M)} \\ &\quad + \cdots \\ &\quad + S_M \frac{(\delta_t - \delta_1)(\delta_t - \delta_2) \cdots (\delta_t - \delta_{M-1})}{(\delta_M - \delta_1)(\delta_M - \delta_2) \cdots (\delta_M - \delta_{M-1})}. \end{aligned} \quad (4)$$

This expression represents a polynomial in δ_t , whose maximum possible value is unity.

Step 11: Compute the Optimum Threshold

Set

$$S_t = 1$$

in the obtained interpolation polynomial and solve for the roots of the equation. These roots correspond to possible values of the threshold δ_t .

Step 12: Select the Optimum Root

Substitute each obtained root back into the interpolation polynomial. Choose the root for which the corresponding value of S_t is closest to unity. The selected threshold is denoted by

$$\eta.$$

Thus, η represents the threshold value corresponding to the maximum silhouette value.

Step 13: Update the Optimizing Factor

For every cluster, update the optimizing factor as

$$\beta_m = \frac{\eta}{\delta_m}, \quad m = 1, 2, \dots, M. \quad (5)$$

The updated values of β_m are substituted back into the threshold equation (Step 2) to compute new threshold values. This update improves the silhouette values of the generated clusters.

Step 14: Repeat Until Convergence

Using the updated optimizing factors, repeat Steps 2–14 until the threshold values δ_m converge and the Global Silhouette Index (GSI) reaches its maximum value.

The optimizing factors corresponding to the iteration with the highest GSI are selected, and the algorithm is executed once again using the optimum number of clusters M to obtain the final clustering result.

Based on extensive experimentation and analysis, the number of iterations required for convergence is chosen as ten.

(3.2) Measures of Cluster Quality

Global Silhouette Index (GSI): The GSI allocates a quality measure $s(i)$ (silhouette width) to each sample in cluster X_m . A value nearing 1 implies a well-clustered sample, while negative values hint at potential misclassification.

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

The overall GSI is determined by averaging the silhouette values for all clusters.

$$GSI = \frac{1}{M} \sum_{m=1}^M S_m$$

Partition Index (PI): This index represents the ratio of cluster compactness to separation. A minimized PI value signifies a superior partitioning of the data.

$$PI = \frac{\sum_{m=1}^M \frac{\sum_{j=1}^n (\mu_{jm})^2 \|\bar{x}_j - \bar{c}_m\|^2}{N_m \sum_{k=1}^M \|\bar{c}_k - \bar{c}_m\|^2}}$$

Separation Index (SI): SI evaluates partition validity using minimum-distance separation between clusters. A lower SI indicates well-separated and compact groupings.

$$SI = \frac{\sum_{m=1}^M \sum_{j=1}^n (\mu_{jm})^2 \|\bar{x}_j - \bar{c}_m\|^2}{n \cdot \min_{k,m} \|\bar{c}_k - \bar{c}_m\|^2}$$

4 Results

(4.1) Running the SOC Algorithm on Test Image

The SOC algorithm was applied to a grayscale test image. The algorithm autonomously determined $M = 3$ as the optimal number of clusters.

Higher GSI implies better cluster quality. Lower SI implies better the cluster.

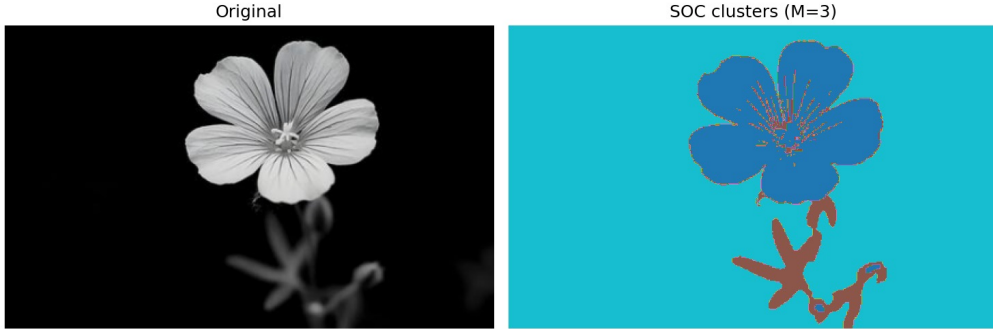


Figure 1: Original grayscale image (left) and SOC segmentation result with $M = 3$ clusters (right).

Table 1 presents the GSI and optimizing factors $\beta_1, \beta_2, \beta_3$ recorded at each of the 10 iterations.

(4.2) Variation of β and GSI across Iterations

The GSI peaks at iteration 1 with a value of 0.6368 and gradually declines, confirming convergence of the optimizing factors.

Table 1: Beta values and GSI per iteration.

Iter	GSI	β_1	β_2	β_3
1	0.6368	1.0000	1.0000	1.0000
2	0.6336	0.8158	0.8557	0.9613
3	0.6365	1.0462	1.0382	1.0000
4	0.6309	0.7669	0.8121	0.9542
5	0.6354	1.1189	1.0972	1.0000
6	0.6247	0.7001	0.7514	0.9489
7	0.6289	1.2365	1.1931	1.0000
8	0.6156	0.6171	0.6758	0.9543
9	0.6085	1.4374	1.3527	1.0000
10	0.5394	0.4990	0.5639	0.9686

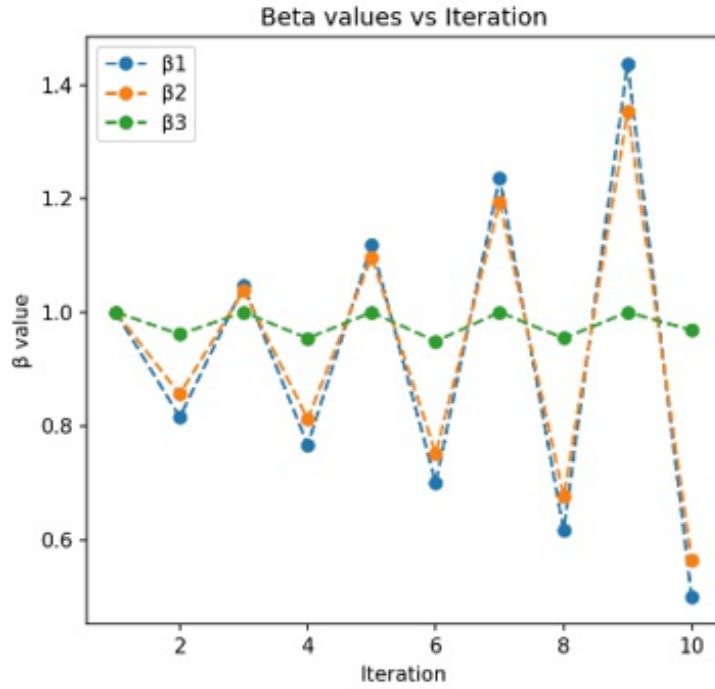


Figure 2: Variation of optimizing factors β_1 , β_2 , β_3 across 10 iterations.

Table 2: Comparison of clustering techniques ($M = 3$). Higher GSI and lower PI, SI indicate better clustering.

Technique	GSI	PI	SI	DI
IMC ($M = 3$)	0.6368	0.0661	0.1512	0.5300
SOC ($M = 3$)	0.6368	0.0661	0.1512	0.5300
FCM ($M = 3$)	0.5745	0.0753	0.1165	0.7442
K-means ($M = 3$)	0.5724	0.0754	0.1161	0.7241

(4.3) Comparison with Other Clustering Techniques

Table 2 compares SOC against IMC, FCM, and K-means for $M = 3$.

SOC matches IMC on all indices while autonomously determining M without requiring it as a prior input, demonstrating its self-optimal nature. Both outperform FCM and

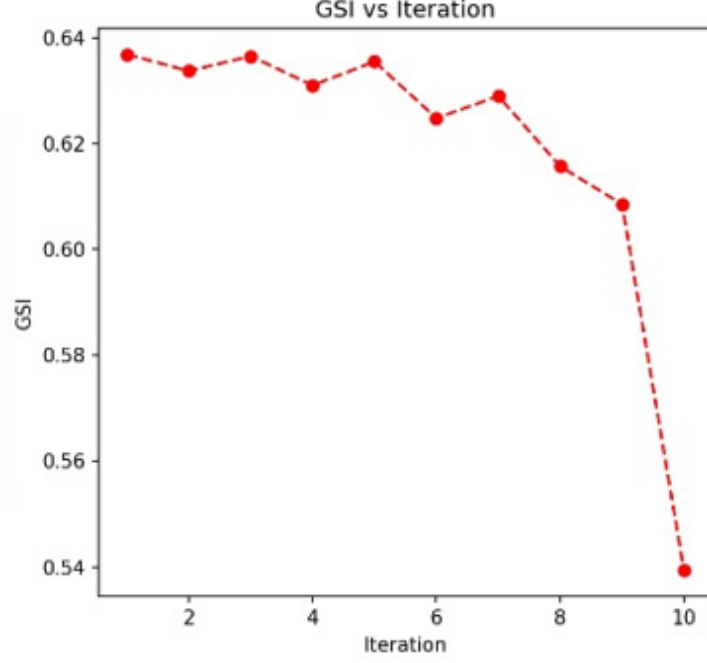


Figure 3: Global Silhouette Index (GSI) across 10 iterations. Best GSI = 0.6368 at iteration 1.

K-means on GSI and PI metrics.

5 Conclusion

The Self-Optimal Clustering (SOC) method is a density-based, parameter-free algorithm that smartly computes both the ideal threshold for forming clusters and the optimum number of clusters itself.

It successfully generates well-separated, high-quality clusters across various testing environments. SOC finds extensive application in fields such as bioinformatics, pattern recognition, sensor networks, and data mining. Its robust performance, which is validated through visual and statistical assessments, establishes it as a highly capable alternative to traditional clustering strategies, particularly when the underlying structure of the data is unknown.